

Latex & Markdown

Introdução Engenharia Informática

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December 8, 2025

Exercises

This guide will take you from setting up your environment to write using latex and markdown to building a web service that automatically converts Markdown files to PDF. No prior coding experience is required—just follow the steps and copy the code blocks exactly.

Step 0: System Setup

First, we need to install the “Compiler” tools.

1. Open your **Terminal**.
2. Copy and paste the following command (you will need your password):

```
sudo apt update
sudo apt install wget texlive-luatex texlive-latex-recommended \
texlive-latex-extra texlive-fonts-recommended texlive-fonts-extra \
fonts-lmodern fonts-noto fonts-noto-cjk fonts-noto-color-emoji pandoc -y
```

3. Create a folder named ex12 to place the exercises.
4. Run this in your terminal inside the ex12 folder first:

```
wget -O profile.png "https://placeholder.co/400x400/png"
wget -O graph.png "https://placeholder.co/600x400/png"
```

Note: texlive-full is the complete version of the texlive tool, but it is large (several GBs). It contains every package you might need so you don't encounter missing errors later. In case of error consider installing it.

Step 1: Write a CV

We will create a CV using both methods. Create a folder named ex12 on your Desktop to keep things organized.

Option A: The LaTeX Way (Professional)

LaTeX has a famous class called moderncv. It looks great out of the box.

1. Create a file named my_cv.tex inside your ex12 folder.
2. Paste this code into it:

```
\documentclass[11pt,a4paper,sans]{moderncv}
\moderncvstyle{classic} % Options: casual, classic, banking, oldstyle, fancy
\moderncvcolor{blue}    % Options: blue, orange, green, red, purple, grey
\usepackage{graphicx} % Required for images

% Personal Data
\name{John}{Doe}
\title{Computer Science Student}
\address{123 University Ave}{Aveiro, Portugal}
\phone[mobile]{+351 912 345 678}
```

```

\email{john.doe@student.pt}

% Profile picture (height: 64pt, border: 0.4pt)
\photo[64pt][0.4pt]{profile.png}

\begin{document}
\makecvtitle

\section{Education}
\cventry{2023--2026}{BSc in Computer Science}{University of Aveiro}
{Aveiro}{}{Focus on Algorithms and Web Development.}

\section{Skills}
\cvitem{Languages}{Python, C, Java, SQL}
\cvitem{Tools}{Git, Docker, LaTeX, Linux}

\section{Experience}
\cventry{2022}{Summer Intern}{Tech Corp}{Lisbon}
{}{Helped fix bugs in the frontend application.}

\end{document}

```

3. **Compile it:** Run this command in your terminal inside the folder:
`pdflatex my_cv.tex`
4. Check the folder. You now have `my_cv.pdf`.

Option B: The Markdown Way (Fast)

1. Create a file named `my_cv.md`.
2. Paste this code:

```

# John Doe
![[Profile Picture](https://placeholder.co/150x150/png)]

**Computer Science Student**
*Aveiro, Portugal | +351 912 345 678 | john.doe@student.pt*

---

## Education
**BSc in Computer Science** | University of Aveiro | *2023--2026*
* Focus on Algorithms and Web Development.

## Skills
* **Languages:** Python, C, Java, SQL
* **Tools:** Git, Docker, LaTeX, Linux

## Experience
**Summer Intern** | Tech Corp | *2022*
* Helped fix bugs in the frontend application.

```

3. **Compile it:**
`pandoc my_cv.md -o my_cv_markdown.pdf`

Step 2: Write a Project Report

Scenario: You are writing a report on “The Efficiency of Coffee on Student Coding Speed”.

Option A: The LaTeX Way

1. Create a file named `report.tex`.
2. Paste this code:

```
\documentclass[12pt]{report}
\usepackage[utf8]{inputenc}
\usepackage{graphicx}
\usepackage{hyperref}
\usepackage{booktabs}

\title{The Efficiency of Coffee on
Student Coding Speed}
\author{Group 12: John Doe \& Jane Smith}
\date{\today}

\begin{document}

\maketitle
\tableofcontents

\chapter{Introduction}
This study analyzes whether caffeine intake
correlates with lines of code written per hour.
\begin{figure}[!htb]
  \centering
  \includegraphics[width=0.8\textwidth]{graph.png}
  \caption{Projected Coding Speed vs. Caffeine}
  \label{fig:coffee_graph}
\end{figure}

\chapter{Methodology}
We observed 10 students over 4 hours.
\begin{table}[!htb]
  \centering
  \caption{Participant Group Settings}
  \label{tab:groups}
  % Notice the @{} to remove side spacing
  \begin{tabular}{@{}llrr@{}}
    \toprule
    Group & Drink Type & Dosage (mg) & \\
    & Participants & \\
    \midrule
    A & Water & 0 & 5 \\
    B & Espresso & 120 & 5 \\
    C & Energy Drink & 160 & 5 \\
    \bottomrule
  \end{tabular}
\end{table}

\chapter{Results}
Group B wrote code 20\% faster but introduced
10\% more bugs.

\chapter{Conclusion}
Coffee increases speed but decreases accuracy.

\end{document}
```

3. **Compile it:**
`pdflatex report.tex`

```
pdflatex report.tex
```

(Note: We run it twice so LaTeX can generate the Table of Contents numbers correctly).

Option B: The Markdown Way

1. Create a file named `report.md`.
2. Paste this code:

```
---
title: The Efficiency of Coffee on Student Coding Speed
author: Group 12 - John Doe & Jane Smith
date: 2023-12-05
---

# Introduction
This study analyzes whether caffeine intake
correlates with lines of code written per hour.

![[Projected Coding Speed vs. Caffeine](graph.png){ width=80% }

# Methodology
We observed 10 students over 4 hours.

Table: Participant Group Settings

| Group | Drink Type | Dosage (mg) | Participants |
| :--- | :--- | ---: | ---: |
| A | Water | 0 | 5 |
| B | Espresso | 120 | 5 |
| C | Energy Drink | 160 | 5 |

# Results
Group B wrote code 20% faster but introduced 10% more bugs.

# Conclusion
Coffee increases speed but decreases accuracy.

3. Compile it (Using the --number-sections flag to make it look like a report):
    pandoc report.md -o report_markdown.pdf --number-sections
```

Step 3: Write Presentation Slides

Now you need to present the findings from Step 2.

Option A: The LaTeX Way (Beamer)

1. Create a file named `slides.tex`.
2. Paste this code:

```
\documentclass{beamer}
\usetheme{Madrid}
\usepackage{booktabs} % Required for the table

\title{Coffee vs. Code}
\author{Group 12}
\date{\today}

\begin{document}

\frame{\titlepage}
```

```

\begin{frame}
\frametitle{Introduction}
\begin{itemize}
\item Does caffeine help us code?
\item We tested 10 students.
\end{itemize}
\begin{center}
\includegraphics[width=0.6\textwidth]{graph.png}
\end{center}
\end{frame}

\begin{frame}
\frametitle{Methodology}
\begin{table}
\centering
\begin{tabular}{@{}llrr@{}}
\toprule
Group & Drink & Dosage & N \\
\midrule
A & Water & 0mg & 5 \\
B & Espresso & 120mg & 5 \\
\bottomrule
\end{tabular}
\end{table}
\end{frame}

\begin{frame}
\frametitle{Results}
\alert{Result:} Speed increased by 20\%.
\end{frame}

\end{document}

```

3. Compile it:

```
pdflatex slides.tex
```

Option B: The Markdown Way (Beamer via Pandoc)

1. Create a file named `slides.md`.
2. Paste this code (Use `---` to separate slides):

```

---
title: Coffee vs. Code
author: Group 12
date: today
---

# Introduction
* Does caffeine help us code?
* We tested 10 students.



---

# Methodology

| Group | Drink | Dosage | N |
| :--- | :--- | ---: | ---: |
| A | Water | 0mg | 5 |
| B | Espresso | 120mg | 5 |

```

Results

****Result:**** Speed increased by 20%.

3. **Compile it** (We tell Pandoc to output type beamer):

```
pandoc slides.md -t beamer -o slides_markdown.pdf
```

Step 4: The Automated Converter (Docker)

Now, we will build a real web application. A user uploads a Markdown file on a website, the server converts it to PDF using Pandoc, and sends it back.

4.1 Folder Structure

Create a new folder named `converter_app`. Inside it, create two folders: `backend` and `frontend`. Your structure must look exactly like this:

```
converter_app/
├── compose.yml
├── backend/
│   ├── Dockerfile
│   ├── main.py
│   └── requirements.txt
└── frontend/
    ├── Dockerfile
    ├── index.html
    └── nginx.conf
```

4.2 The Files

1. `converter_app/compose.yml` (This tells Docker to run both the Python API and the Web Server).

```
services:
  backend:
    build: ./backend
    ports:
      - "8000:8000"
    volumes:
      - ./backend:/app

  frontend:
    build: ./frontend
    ports:
      - "8080:80"
```

2. `converter_app/backend/Dockerfile` (This installs Python + Pandoc + LaTeX inside the container).

```
FROM python:3.12-trixie
```

```
# Install system dependencies (Pandoc and LaTeX)
```

```
RUN apt update && apt install -y \
    texlive-luatex texlive-latex-recommended \
    texlive-latex-extra texlive-fonts-recommended \
    texlive-fonts-extra fonts-lmodern fonts-noto \
    fonts-noto-cjk fonts-noto-color-emoji pandoc \
    && rm -rf /var/lib/apt/lists/*
```

```
WORKDIR /app
```

```
COPY requirements.txt .
```

```
RUN pip install --no-cache-dir -r requirements.txt
```

```
COPY . .
```

```
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

3. converter_app/backend/requirements.txt

```
fastapi
uvicorn
python-multipart
```

4. converter_app/backend/main.py (The Python logic: Receive file -> Run Pandoc -> Return PDF).

```
import subprocess
import os
from fastapi import FastAPI, UploadFile, File
from fastapi.responses import FileResponse
from fastapi.middleware.cors import CORSMiddleware
```

```
app = FastAPI()
```

```
# Allow the frontend to talk to this backend
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)
```

```
@app.post("/convert")
```

```
async def convert_md_to_pdf(file: UploadFile = File(...)):
```

```
    # 1. Save the uploaded markdown file
```

```
    input_filename = "input.md"
```

```
    output_filename = "output.pdf"
```

```
    with open(input_filename, "wb") as f:
        f.write(await file.read())
```

```
    # 2. Run Pandoc command inside the container
```

```
    # pandoc input.md -o output.pdf
```

```
    try:
```

```
        subprocess.run(
            ["pandoc", "--pdf-engine=lualatex",
             input_filename, "-o", output_filename],
            check=True
        )
```

```
    except subprocess.CalledProcessError:
```

```
        return {"error": "Conversion failed"}
```

```
    # 3. Return the generated PDF
```

```
    return FileResponse(output_filename, filename="converted.pdf", media_type='application/pdf')
```

5. converter_app/frontend/Dockerfile

```
FROM nginx:alpine
```

```
COPY nginx.conf /etc/nginx/conf.d/default.conf
```

```
COPY index.html /usr/share/nginx/html/index.html
```

6. converter_app/frontend/nginx.conf

```
server {
    listen 80;
```

```

server_name localhost;

location / {
    root    /usr/share/nginx/html;
    index  index.html index.htm;
}

```

7. converter_app/frontend/index.html (The User Interface).

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Markdown to PDF Converter</title>
  <style>
    body { font-family: sans-serif; display:
    flex; justify-content: center; align-items: center;
    height: 100vh; background: #f0f0f0; }
    .card { background: white; padding: 2rem;
    border-radius: 10px; box-shadow: 0 4px 6px rgba(0,0,0,0.1);
    text-align: center; }
    button { background: #007bff; color: white; border: none;
    padding: 10px 20px; border-radius: 5px; cursor: pointer;
    margin-top: 10px; }
    button:hover { background: #0056b3; }
  </style>
</head>
<body>
  <div class="card">
    <h1>MD to PDF Converter</h1>
    <p>Select a Markdown file to convert.</p>
    <input type="file" id="fileInput" accept=".md">
    <br><br>
    <button onclick="uploadAndConvert()">Convert & Download</button>
    <p id="status"></p>
  </div>

  <script>
    async function uploadAndConvert() {
      const fileInput = document.getElementById('fileInput');
      const status = document.getElementById('status');

      if(fileInput.files.length === 0) {
        alert("Please select a file!");
        return;
      }

      const formData = new FormData();
      formData.append("file", fileInput.files[0]);
      status.innerText = "Converting... Please wait.";

      try {
        // Send file to Backend (FastAPI)
        const response = await fetch('http://localhost:8000/convert', {
          method: 'POST',
          body: formData
        });

        if (!response.ok) throw new Error("Conversion failed");

        // Create a blob from the response (the PDF)

```

```

const blob = await response.blob();
const url = window.URL.createObjectURL(blob);

// Force download
const a = document.createElement('a');
a.href = url;
a.download = "converted_document.pdf";
document.body.appendChild(a);
a.click();
a.remove();
status.innerText = "Done!";
} catch (error) {
  console.error(error);
  status.innerText = "Error converting file.";
}
}
</script>
</body>
</html>

```

4.3 Running the App

1. Open your terminal inside the `converter_app` folder.
2. Run the following command:

```
sudo docker compose up --build
```

(This will take a few minutes to download the Python image and install LaTeX inside it. Be patient).

3. Once it stops moving and says "Application startup complete":
 - Open your browser (Firefox/Chrome).
 - Go to: `http://localhost:8080`
4. Upload the `my_cv.md` or `report.md` you created earlier.
5. Click **Convert**. The browser should download a PDF version of your Markdown file!